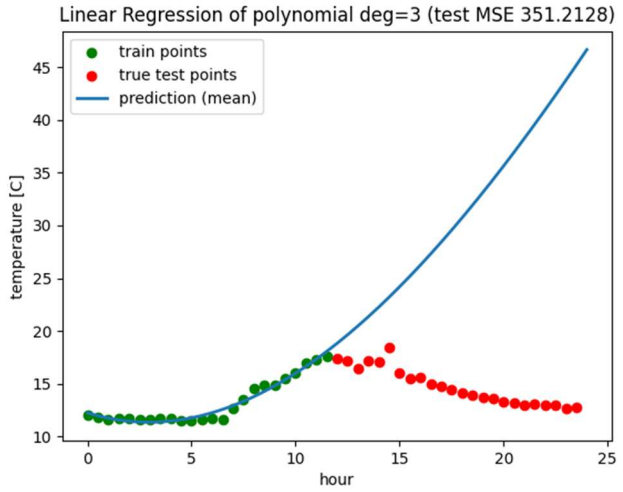
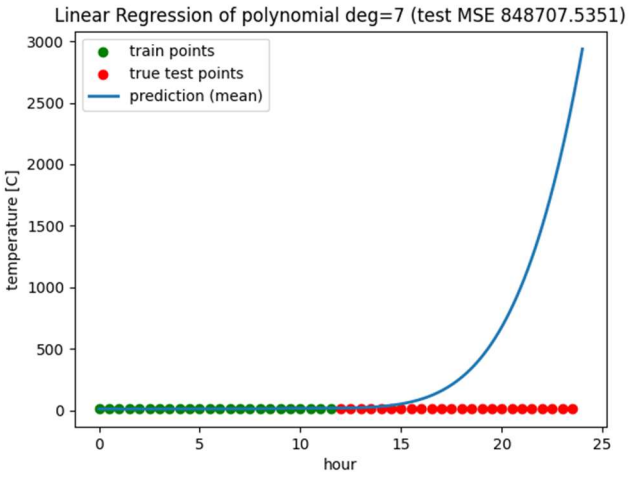
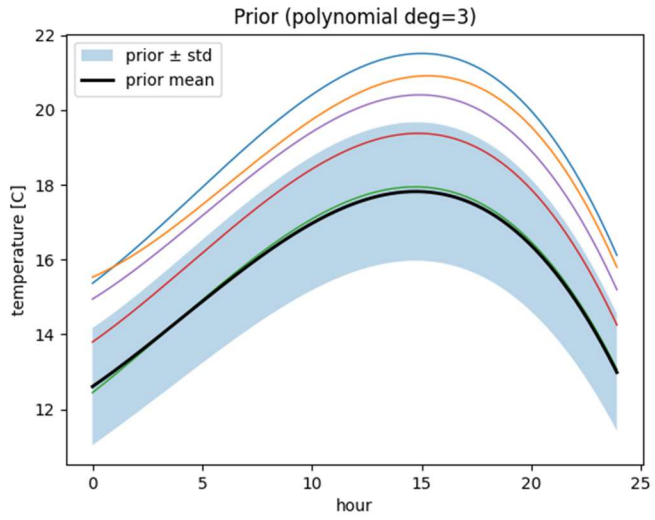
Polynomial Basis Functions

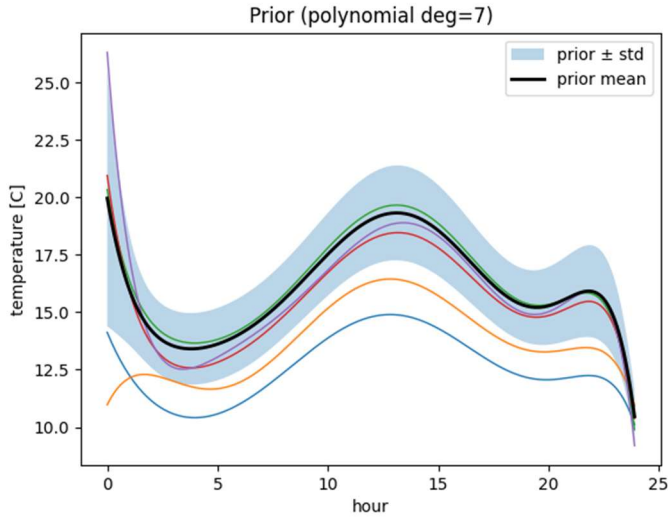
3) Fitting a linear regression model to the first half of November 16th. This is the plot of the true test points together with the predicted curve.



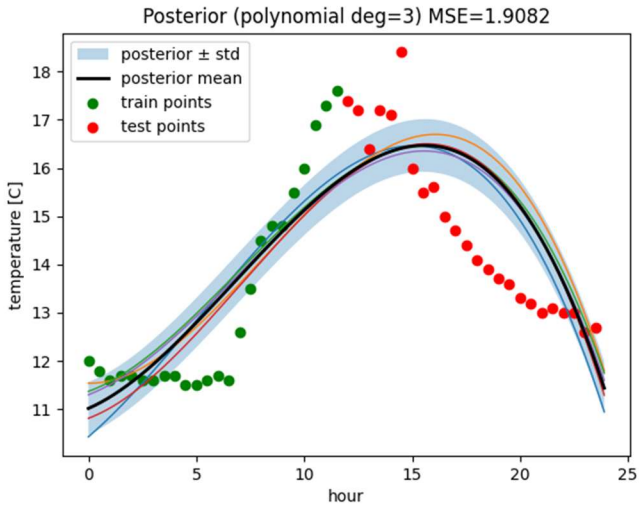


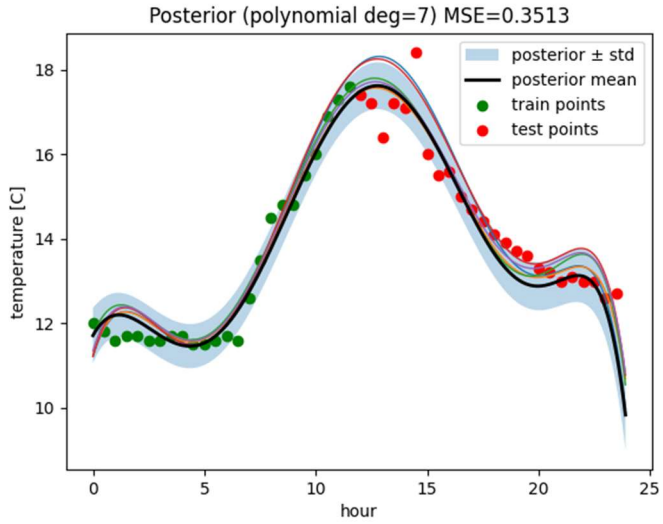
5) Using historical data, we try to learn a Gaussian prior over θ .





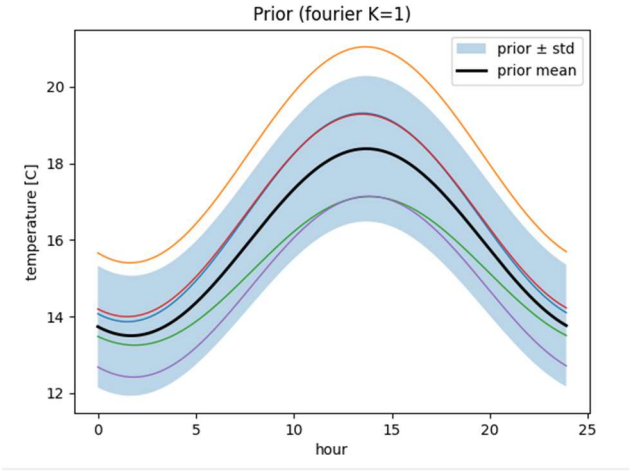
6) Fitting a BLR model to the first half of the temperatures to get the posterior and predict the

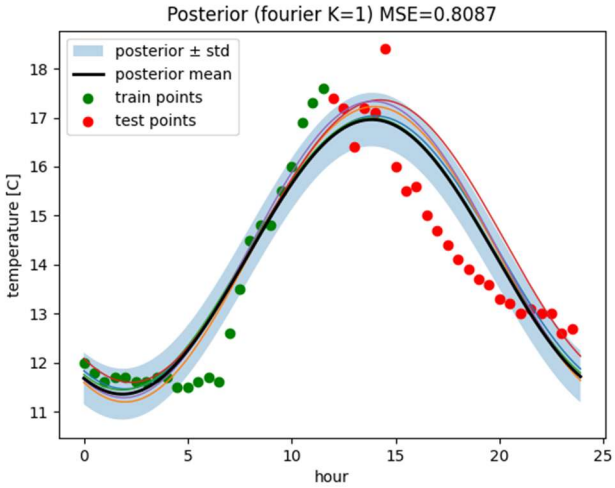




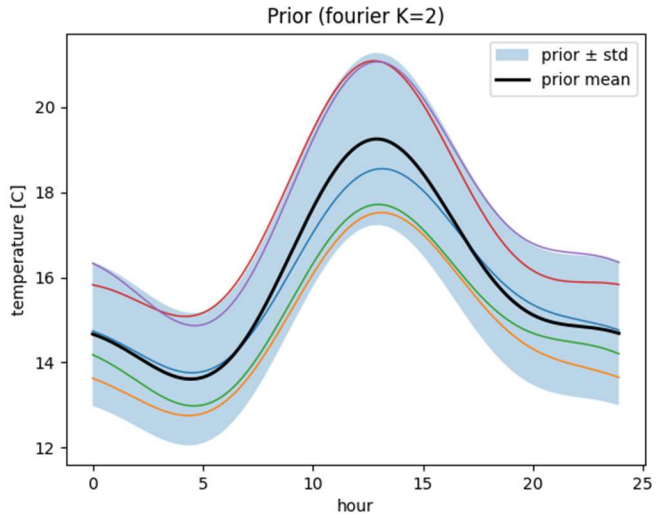
Fourier Basis Functions

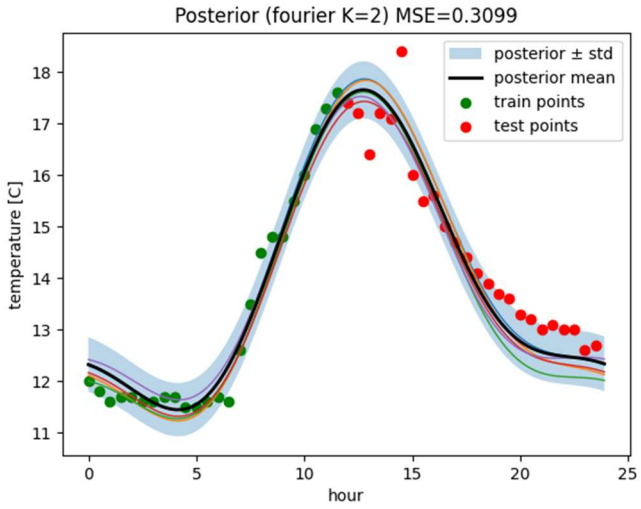
For K=1:



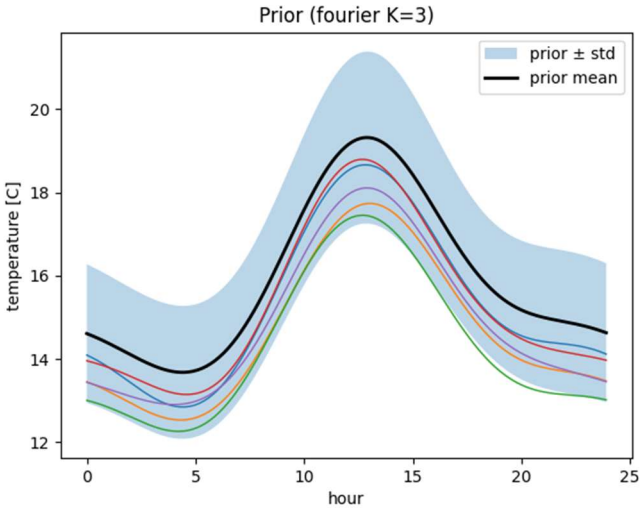


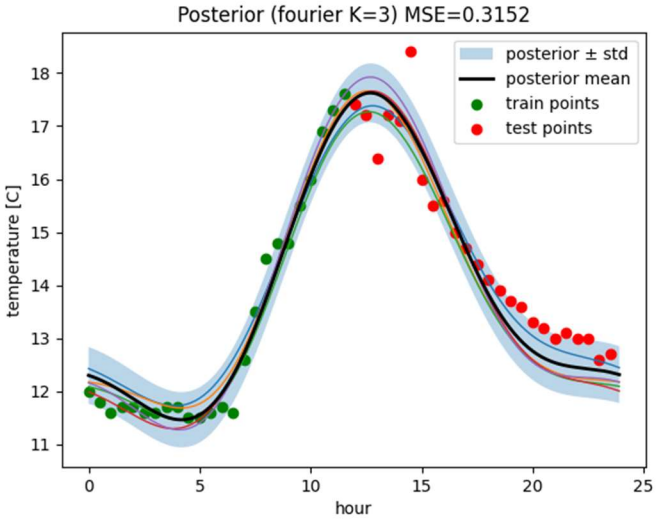
For K=2:


Prior (fourier K=2)
prior ± std
prior mean
temperature [C]
hour

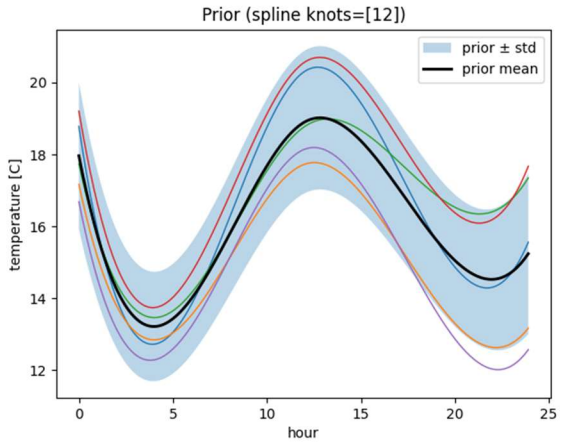

Posterior (fourier K=2) MSE=0.3099
posterior ± std
posterior mean
train points
test points
temperature [C]
hour

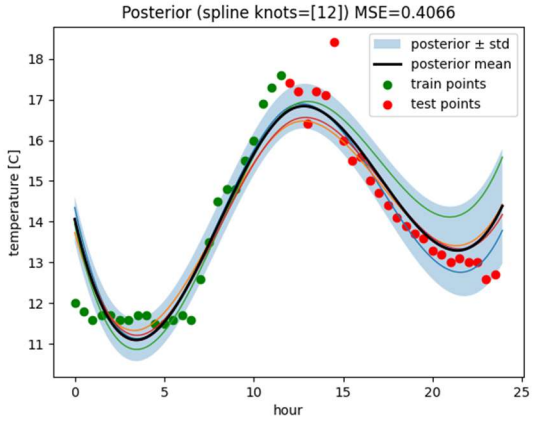
For K=3:

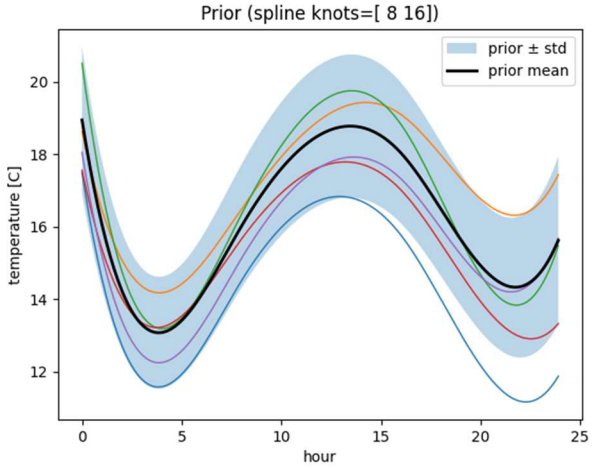

Prior (fourier K=3)
prior ± std
prior mean
temperature [C]
hour

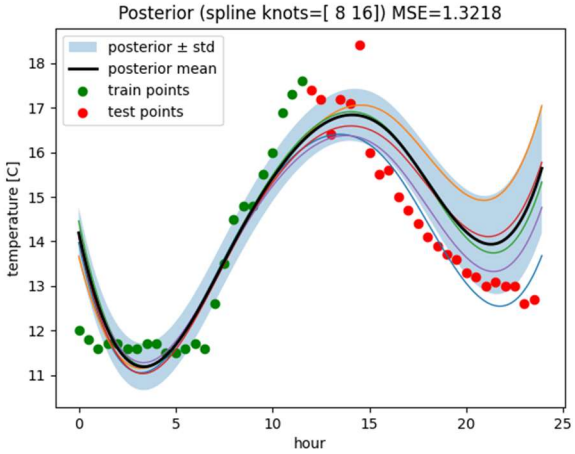

Posterior (fourier K=3) MSE=0.3152
posterior ± std
posterior mean
train points
test points
temperature [C]
hour

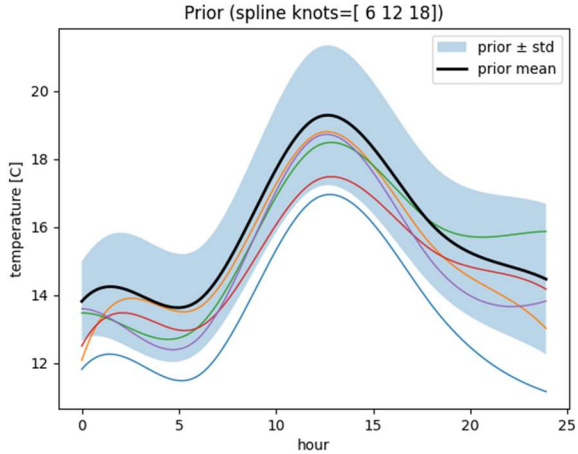
Cubic Regression Spline

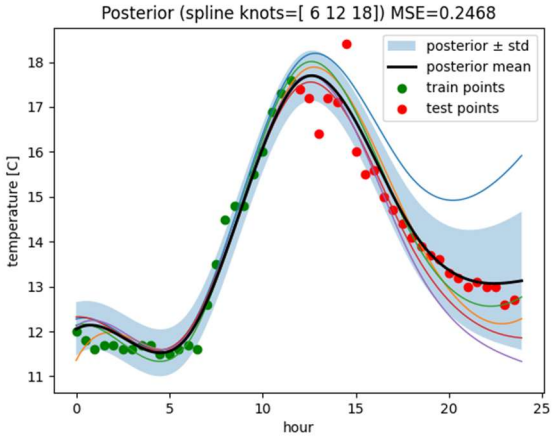












All the results for the three different kinds of basis functions:

```
Average squared error with LR and polynomial degree=3: 351.2128
Average squared error with LR and polynomial degree=7: 848707.5351
Bayesian LR (polynomial deg=3) MMSE test MSE: 1.9082
Bayesian LR (polynomial deg=7) MMSE test MSE: 0.3513
Bayesian LR (fourier K=1) MMSE test MSE: 0.8087
Bayesian LR (fourier K=2) MMSE test MSE: 0.3099
Bayesian LR (fourier K=3) MMSE test MSE: 0.3152
Bayesian LR (spline knots=[12]) MMSE test MSE: 0.4066
Bayesian LR (spline knots=[ 8 16]) MMSE test MSE: 1.3218
Bayesian LR (spline knots=[ 6 12 18]) MMSE test MSE: 0.2468
```